

BACHELOR THESIS
Dion Osmani

Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Informatique et Systèmes de communication (ISC)

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Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
Haute École d'Ingénierie de Sion

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Abstract

The abstract of a bachelor thesis should provide a concise summary of the entire work. It typically includes:

- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear, and usually does not exceed 250–300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

Please also insert your project git/github URL HERE if your project is not confidential.

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Keywords : engineering, data, large language models, AI agents, energy planning

Résumé

Le résumé d'un mémoire de bachelor doit fournir un aperçu concis de l'ensemble du travail. Il inclut généralement :

- Le contexte et la motivation de la recherche.
- L'objectif principal ou la question de recherche.
- Une brève description de la méthodologie ou de l'approche utilisée.
- Les principaux résultats ou découvertes.
- La conclusion principale ou les implications du travail.

Le résumé doit être autonome, clair et ne pas dépasser habituellement 250 à 300 mots. Il permet aux lecteurs de comprendre rapidement le but et les résultats du mémoire sans lire l'intégralité du document.

Le résumé doit être rédigé en français **et** en anglais.

Veuillez également ajouter l'URL de votre git/github ici si le projet n'est pas confidentiel.

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Keywords : engineering, data, large language models, AI agents, energy planning

Acknowledgements

The **Acknowledgements** section of a bachelor thesis is where you express gratitude to those who supported you during your research and writing process. It is an **OPTIONAL** section. It may include:

- Academic supervisors or advisors who provided guidance.
- Professors or instructors who offered feedback or resources.
- Family and friends for emotional or practical support.
- Institutions or organizations that provided funding, facilities, or data.
- Anyone else who contributed significantly to your work.

Keep this section concise and sincere. It is typically placed after the abstract and before the main content of your thesis.

Table of Contents

1 – Writing a thesis	13
1.1 – The content of a thesis	13
2 – Introduction	15
3 – State of the Art	17
4 – Methodology	19
4.1 – Requirements	19
4.2 – System Design	21
5 – Results	27
6 – Discussion	29
7 – Conclusion	31
Appendix	35

Chapter 1 – Writing a thesis

Writing a report is an exercise that involves both **content and form**. In this document, we aim to simplify the formatting aspect without making any assumptions about the content, specifically in the context of the ISC program¹.

1.1 – The content of a thesis

The general structure of a bachelor thesis typically includes the following sections:

1. **Abstract:** A concise summary of the thesis, including the research question, methodology, results, and conclusions.
2. **Résumé:** A summary of the thesis in French.
3. **Acknowledgements:** (Optional) A section to thank those who supported the work.
4. **Table of Contents:** An organized listing of chapters and sections.
5. **Introduction:** Presents the background, motivation, objectives, and scope and plan of the thesis.
6. **State of the Art / Literature Review:** Reviews existing research and situates the thesis within the academic context. If salient in your work.
7. **Methodology:** Describes the methods, materials, and procedures used in the research / thesis.
8. **Results:** Presents the findings of the research, often with tables, figures, and analysis.
9. **Discussion:** Interprets the results, discusses implications, and relates findings to the research question.
10. **Conclusion:** Summarizes the main findings, contributions, and suggests future work.
11. **References / Bibliography:** Lists all sources cited in the thesis.
12. **Appendices:** (Optional) Contains supplementary material such as raw data, code, or additional explanations.

This structure may vary depending on the field of study, but these elements are commonly found in most bachelor theses. They are compulsory for the *ISC Bachelor thesis*.

¹Here is how to add a footnote <https://isc.hevs.ch>

Chapter 2 – Introduction

TODO: diviser en sections? contexte, problématique...

Over the past few decades, society has been sensitized and slowly became more aware of significant problems that we are likely to face in the coming years.

Climate change and other environmental issues arise as a result of human-driven activities.

Scientists have monitored this matter and proposed various frameworks to address and mitigate these problems. In Switzerland, these different frameworks are implemented in the legislation and guidelines (at federal and canton levels) to steer the country towards a more sustainable future. Municipalities may introduce in their regulations energy requirements that are more constraining than those set by the cantonal law. TODO: citer loi teams jessen?

Municipalities in Switzerland are required to submit an energy planning document which outlines their future strategies to comply with those directives while also considering the characteristics of their energetical landscape.

These different properties can be quantified and analyzed through the use of a very valuable resource: data. Data is emitted by various sources ; sensors, energy models or citizen records for e.g. all yield datapoints that help us assess different indicators we are willing to measure against our municipality. These indicators, *in fine*, help us evaluate our progress towards energy-related goals.

Over the past few years, Artificial Intelligence (AI) has rapidly transformed our habits when interacting with information.

Large Language Models (LLM) allow users to interact with these systems in natural language facilitating the interface between humans and *machines*. They can provide insights into vast amounts of data at speed and scales which are beyond our capabilities.

This work tackles this exact problem that is the implementation of a solution which assists users into energy planning for a municipality.

This complex problem is approached by leveraging the power of *specialized* AIs which each offer expertise into a variety of domains and are coordinated using an *orchestration* AI to provide a solution. The expertise of the user interacting with the system tailors this solution to the specificities of the municipality.

The key steps in the engineering of this implementation are identifying the key information and datasources that are relevant to this process, structure the different AIs into an architecture whose components and interfaces are well-defined and ultimately implementing the solution.

The main objective of this work is to investigate how effective and reliable such a solution is and to assess its strengths and weaknesses. Additional goals are also outlined:

- Defining the important information in assisting user decision making.
- Understanding which datasources are available and relevant for this decision making.
- Structuring the decision by using an orchestration AI and many specialized AIs.
- Training specialized AIs on specialized datasets.
- Simplifying the user interface by handling the communication between the orchestration AI and the specialized AIs.

The project is scoped to municipalities within the canton of Valais/Wallis and strictly relies on publicly available data. Certain measures are taken to ensure the privacy and security of data that would not be of public order considering the future implementation of extra datasources.

The solution is designed to offer a user-friendly interface from which users can interact with the system in a conversational manner and visualize a map of the municipality with different layers.

User behaviour is analyzed to takeaway user preferences which gradually adapt the answers to better meet the user expectations.

TODO: provide a brief overview of the structure of the thesis (plan), add reference to extra scope in methodology?

Chapter 3 – State of the Art

Large language models are highly effective tools for natural language processing and offer various opportunities to enhance our day-to-day tasks and workflows. Ever since they have been introduced to the public, they have been adopted across a wide range of fields and applications.

TODO: citer proprement

The AI Institute at ITMO University published in 2025 a paper titled *LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems*. The study examines how the natural language processing strengths of LLMs, combined with the distributed problem-solving abilities of multi-agent systems, can enhance urban decision-making processes.

The research focused on the testing of three hypotheses: (1) evaluating the capability of LLM agents to effectively route and process diverse urban queries against existing urban information systems, (2) the effectiveness of Retrieval-Augmented Generation (RAG) technology in improving response accuracy when working with local knowledge and regulations and (3) the impact of integrating LLM agents with existing urban information-systems - increasing efficiency and decreasing the decision making process time.

LLM agents (sometimes called AI agents) are software systems that use AI to pursue goals and complete tasks on behalf of users. They show reasoning, planning, and memory and have a level of autonomy to make decisions, learn, and adapt as per cloud.google.com. **mieux citer??**

Their proposed solution was tested against 150 question-answer pairs and used St. Petersburg's Digital Urban Platform as a testbed. The testing dataset was curated and built by a group of human experts such as specialists in urban data analysis, GIS specialists, and urban architects.

They then evaluated different configurations of LLM agents and state-of-the-art models against two primary metrics: G-eval and answer relevance. The G-eval metric provides greater compliance with human requirements as it uses LLMs to evaluate answers from other LLMs based on custom user criteria. These criteria can for e.g. be provided as a list of rules specifying precise steps the LLM should take for evaluation, mirroring human reasoning process. The answer relevance metric, on the other hand, assesses the relevance of the answer from the LLM when compared with the correct answer provided by the experts. This process also leverages LLMs as it first extracts the different statements from the answer and then compares those to the reference answer. Both metrics are bounded between 0 and 1, where a higher value indicates better performance.

In summary, the results show greater performance when integrating the RAG technology and urban information-systems to the solution (G-eval scores of 0.68-0.74) compared to standalone LLM responses (G-eval scores of 0.30-0.38). Relevance scores, on the other hand, remain high whatever the configuration as they are inherently designed to produce semantically relevant text. They also concluded that this research proved practical real-world city management application as it enables efficient processing of urban planning tasks while maintaining high relevance in responses and shortening task completion time from days to hours.

The ITMO study presents a research-driven implementation of LLM agents focusing on decision support through integration with urban data platforms which curate and process urban data to provide insights and recommendations for urban planning and management. Rather than relying on these large-scale platforms, the present thesis explores the potential of leveraging publicly available data from federal and cantonal sources while also considering the interface with municipal archives and residents' files. This work strongly values the user experience with efforts in enhancing conversational interactions through preference-driven reporting and improving the clarity and quality of reported decisions. It neither serves as a continuation nor a re-implementation of the ITMO study, but rather represents an independent application of AI agents to a related use case specifically adapted to the context of a Bachelor's thesis and shaped by my practical implementation choices and problem-solving approach. Any solution designed and implemented around

similar goals and data-related constraints, regardless of the specificities of the use case, may result in an architecture that is somewhat similar.

As we forget about the use case and consider more generic research on the matter, we encounter publications that rather focus on the optimization of various aspects of AI agents, such as their scalability, efficiency, and robustness. While this work tackles some of these challenges, it does not incorporate major research efforts into these areas.

TODO: citer aino différemment? Another AI-centric solution relevant to this use case is [aino](#), described on their homepage as an *AI GIS Analyst for Urban planning teams*. Developed and marketed in the United States, it is a commercial business solution offering a platform where an AI analyzes sites and provides visual insights from simple questions. This solution inspired me into a few design and user experience improvements in my work as no implementation details are provided.

These two points of view position this work within the fast-changing field of AI-driven solutions for urban planning and beyond.

Chapter 4 – Methodology

The following chapter provides an overview of the methodology that has been adopted in this work and describes the approach taken to design, implement and further evaluate the proposed solution. A clear emphasis is put onto the key decisions that have shaped the solution throughout its development as this chapter aims to offer full transparency and reproducibility which enables readers to understand the rationale and structure behind it.

4.1 – Requirements

Identifying and describing the primary requirements lays the groundwork for the design of the solution. The main requirements were established and further refined as the project progressed to meet the expectations and project goals throughout ongoing discussions with the supervisors. Those are categorized into functional and non-functional requirements. Functional requirements focus on user requirements and product features whereas non-functional requirements focus on user expectations and product properties.

The first step is to understand the problem that is solved. Energy planning typically involves:

- Identifying available energy resources².
- Characterizing the needs.
- Assessing different measures and their impacts ; sobriety (reducing energy consumption), efficiency (more efficient technologies) and production of renewable energy sources.

Hence, assisting users in energy planning requires a solution that can gather relevant data sources, analyze and present the current energy landscape of the municipality and provide actionable recommendations tailored to the context of the municipality while ensuring compliance with the legislation and guidelines that apply to said municipality.

Besides that, it had been requested that the user interface has a map showcasing the assessed datapoints within the municipality as well as for the AI to be able to *remember* the user's preferences and past interactions to have the answer better fit what the user expects.

These initial requirements established the basis for the project. As the solution was evaluated with supervisors on a weekly basis, additional requirements emerged gradually shaping the solution to enhance the overall solution and meet the needs of energy planning.

TODO: revoir la dernière phrase, ajouter des autres requirements ?

These requirements are summarized in the Table 1 below:

²The resources and needs are assessed within the geographical boundaries of the municipality, only.

Requirement	Type	Description
Gather relevant data sources	Functional	The solution must collect and integrate data from various sources relevant to the municipality's energy landscape.
Analyze and present energy landscape	Functional	The system should process and visualize the current energy situation of the municipality, enabling users to understand available resources and needs.
Provide actionable recommendations	Functional	The solution must generate tailored recommendations for energy planning, considering the specific context and characteristics of the municipality.
Interactive map interface	Functional	The user interface should include a map displaying assessed datapoints within the municipality for better spatial understanding.
Conversational interface	Functional	The system should provide a natural language interface allowing users to interact with the AI through conversational queries.
Preference memory	Functional	The AI should remember user preferences and past interactions to adapt responses and improve user experience over time.
Multi-AI orchestration	Functional	The system should coordinate multiple specialized AIs through an orchestration layer to provide comprehensive energy planning assistance.
Usability and accessibility	Non-functional	The interface should be intuitive and accessible for users with varying technical expertise.
Data privacy and security	Non-functional	The solution must ensure the privacy and security of any non-public data, especially considering future integration of additional datasources.
Extensibility and adaptability	Non-functional	The architecture should allow for the addition of new features and datasources as requirements evolve.

Table 1 - Requirements table

TODO: ajouter nice to have multilingue, persistance, ...

4.2 – System Design

Renommer implémentation ?

Parler de tech stack, définition des données ?

A modular, scalable and adaptable architecture was designed to ensure that the solution can adapt to evolving requirements and facilitate future enhancements. The system is organized into distinct components which are all responsible for a specific set of functionalities and ensure clear separation of concerns and ease of maintenance.

The following chapter covers both the design and the implementation aspects of the solution.

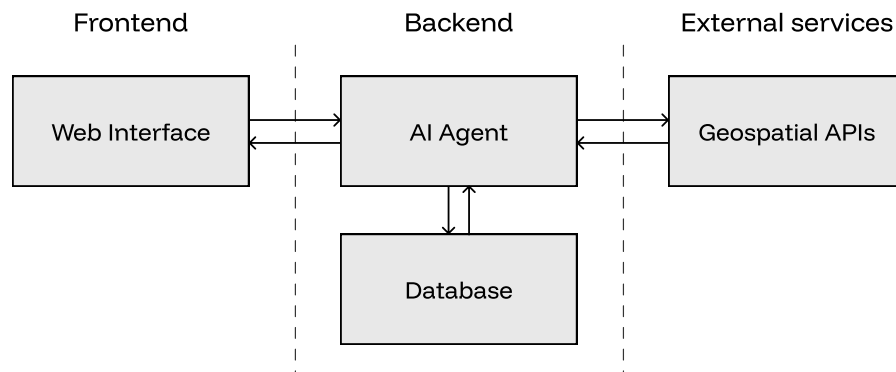


Figure 1 - Global system design

TODO: mettre des numéros sur la figure?? mettre des légendes ? TODO: ajouter un label aux flèches?

The global architecture in its most simplified form is presented in the Figure 1 above. The system is broken down into three distinct layers:

- The frontend layer manages user interaction and presentation of data, providing an intuitive interface for users to communicate with the system.
- The backend layer is responsible for business logic, orchestrating AI agents, processing data, managing a database and handling requests from the frontend.
- The external services layer provides access to third-party Application Programming Interfaces (APIs), a set of protocols and tools that allows different software components to communicate with each other, enabling the system to retrieve data from external platforms and services.

TODO: spécifier que database est redis ?? The layers are made up of various components. The basic dataflow between them is presented in the Table 2 below:

From	To	Data
User (Website)	AI Agent	Prompt/query
AI Agent (Backend)	Local Database	Data retrieval request
AI Agent (Backend)	Third-party APIs	API data request
Local Database	AI Agent (Backend)	Relevant data
Third-party APIs	AI Agent (Backend)	API response data
AI Agent (Backend)	User (Website)	Streamed response

Table 2 - Global system dataflow

The nature and type of data that is exchanged between the AI agent and both the local database and third-party APIs will be later discussed.

Having established a high-level understanding of the system's architecture, the following sections delve into the internal details of each component, starting with the heart of the solution: the AI agent.

4.2.1 – AI agent

In recent months, AI agents have gained traction with the rapid advancement of AI technologies and the increasing demand for personalized and intelligent services. As a result, the term “AI agent” has become a buzzword, with product designers frequently applying the label to a wide range of technologies.

Most definitions agree that the key behaviour that distinguishes AI agents from other solutions is their degree of autonomy as they are able to operate and make decisions independently to achieve a set goal. The complete solution whose sole task is urban energy planning aligns with this definition, as do each of its individual components. Therefore, the term “AI agent” will be used to describe both the entire system and its subsystems.

Human conversations rely on context and prior knowledge and so does the system’s architecture. To deliver a conversational experience, it is essential that the architecture is built to effectively preserve and re-use this context throughout the discussion. One might suggest that the straightforward approach to maintaining conversational context is to include all previous exchanges with the user. However, this method can be very inefficient and comes at great cost.

LLMs rely on a self-attention mechanism to identify and concentrate on the most relevant parts of the input sequence. Each token, the basic unit of text (word, single character, group of words...) that is processed by the model, is assigned a weight reflecting its importance. This allows the model to prioritize relevant information and ignore irrelevant details.

Hence, when the entire conversation history is provided to the model, important tokens may get lost in the larger context and potentially lead to incorrect responses (attention diffusion).

Considering this, a more efficient approach is proposed, relying on a single key assumption: each conversation focuses exclusively on energy planning for one municipality at a time. Accordingly, the conversational context is modeled as a single object that is updated at every turn. It is defined in the codeblock Listing 1 :

```

1  class State:
2      messages: list
3      lang: str
4
5      intent: str
6      location: str
7      aggregated_query: str
8      conversation_type: str
9      needs_clarification: bool
10     needs_memoization: bool
11
12     context_tools: dict
13     context_constraints: list

```

Listing 1 - Conversation state

TODO: mettre au format UML?? TODO: mettre les sous-objets, router, ...? TODO: citer pydantic, runtime, tralala???

The different agents leverage Reasoning Language Models (RLMs), a type of LLM designed to tackle problems by breaking them into logical steps, mimicking human reasoning. Compared to standard language models, they are particularly valuable for tasks that require logical deduction and planning but come with notable drawbacks as they are typically more computationally intensive, leading to higher operational costs and increasing latency in response times. TODO: citer correctement acronyme

By constraining the conversational state and narrowing the scope of each agent, it is possible to reduce the computational load and latency by simply swapping out these large reasoning models by smaller, better-suited models. Doing so, it becomes possible to select reasoning models that are better suited to specific tasks while reducing the computational costs.

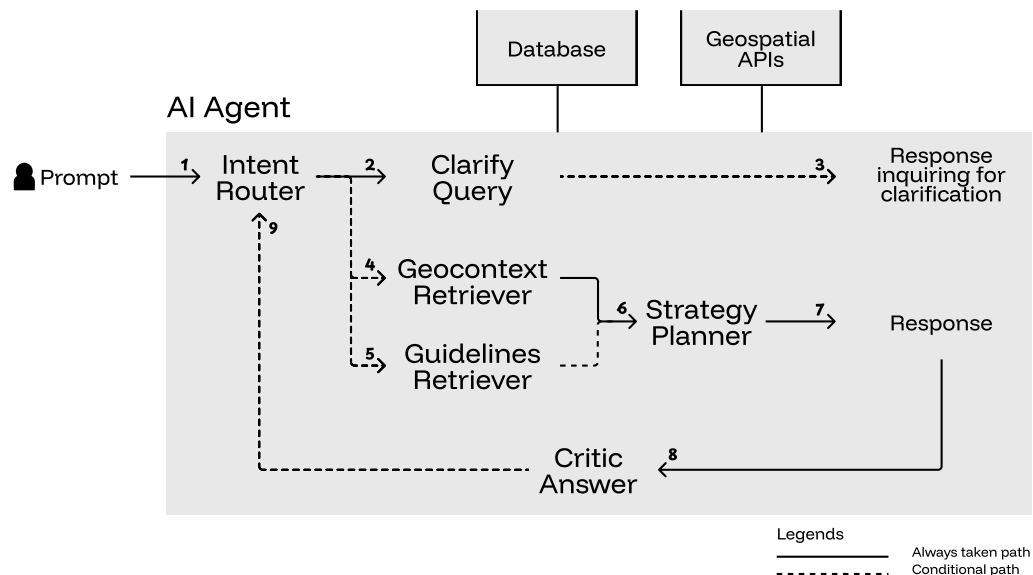


Figure 2 - AI agent architecture

TODO: remplacer par un schéma de FSM classique??

The architecture in the Figure 2 above is modeled after a Finite State Machine (FSM), where each node represents an agent and each edge represents a transition that is either always executed (solid) or conditionally executed (dashed). The dynamic flow of control between agents is guided by the evolving conversational state. It is finite, per definition, as the state takes value in a discrete set. TODO: citer acronyme correctement?

On the implementation-side, LangGraph³, an open-source Python framework, is used to implement the AI agent architecture. Unlike linear pipelines, LangGraph uses a graph abstraction by default, which is particularly well-suited for this state machine architecture. This graph-based structure brings determinism to the system's behavior as the flow between agents is defined by the architecture itself, rather than being dynamically determined by agent-to-agent conversations as in frameworks like Microsoft's AutoGen⁴. Another framework that had been considered was PydanticAI⁵ which offers a structured, type-safe approach to building agent systems by leveraging Pydantic models for inter-agent communication and behaviour definitions. However, it lacks the built-in support for complex state transitions.

All of the available multi-agent AI frameworks are relatively novel and in constant evolution. LangGraph benefits from being built on top of the already renowned LangChain⁶ ecosystem which adds to its reliability and ease of integration with other technologies. Pydantic's type safety will still be implemented within the project to enhance data validation and error handling.

TODO: mettre la techstack dans un chapitre différent??

The main responsibility of each agent is as follows:

- The **Intent Router** routes the user's query to the appropriate agents and accumulates query context.
- The **Clarify Query** clarifies the user's query if it is ambiguous or incomplete.
- The **Geocontext Retriever** retrieves the geospatial data relevant to the analysis.
- The **Guidelines Retriever** retrieves the relevant energy planning guidelines relevant to the analysis.
- The **Strategy Planner** plans the energy strategy based on the retrieved data and guidelines.
- The **Critic Answer** evaluates the proposed energy planning strategy and possibly restarts the whole process.

³<https://www.langchain.com/langgraph>

⁴<https://www.microsoft.com/en-us/research/project/autogen/>

⁵<https://ai.pydantic.dev/>

⁶<https://www.langchain.com/>

With the overall solution defined, the following sections dig in the details of each agent and their implementation. **TODO: plus en détails mgl**

4.2.1.1 – Intent Router

The intent router is a crucial component of the solution. It is the entrypoint of the system and orchestrates the different agents.

Upon receiving a user prompt, the agent analyzes the query to extract its underlying intent. This involves identifying these elements:

- The intent: specifies whether the query is “factual” (for e.g. requesting data) or “actionable” (seeking planning guidance, recommendations, or strategic advice).
- The location: the municipality name mentioned in the user request, if available.
- The aggregated query: a summary that combines all available context from the current conversation and the previous query into a single one.
- The conversation type: identifies the conversational context ; “new_analysis” (fresh query), “correction_request” (user questions the accuracy of a previous response) or ‘follow_up’ (user requests additional detail or expansion on the same topic).
- The need for clarification: defines whether more information is needed to understand what the user wants (e.g., missing location, unclear intent, or vague request).
- The needs for memoization: specifies if the user provided explicit preferences, corrections to assumptions, or scope refinements that should be remembered for future queries (for e.g. the format used to summarize the retrieved data or only considering a single aspect from certain datapoints).

Implementation-wise, the output of the language model is constrained to a single Pydantic data schema, *RouterOutput*. While these models typically generate natural language responses, these complex multi-agent systems benefit from a structured output format that can easily be further processed. This is possible thanks to OpenAI⁷, introducing the support for structured outputs in late 2024, a feature that has since been adopted by many providers. **TODO: inclure prompt system**

All these fields except the aggregated query take value in a finite set of options (considering the *location* field is either set or unset.). Consequently, it is very easy to plan and orchestrate the following actions.

Since the application must offer a conversational experience, the previous *RouterOutput* is accumulated on every turn. Past fields are only updated if they differ from the new ones. As such, context and knowledge is properly accumulated over time. Once the municipality is provided, for example, it is not needed anymore as the request is assumed to concern the same municipality.

On the other hand, when a request treats a different municipality, both the *context_tools* and *context_constraints* defined in Listing 1 are reset. This way, the data associated with the previously discussed municipality is cleared. To ensure correct implementation, whenever a request concerns a different municipality, both the *context_tools* and *context_constraints* defined in Listing 1 are reset. This means the geospatial data and guidelines previously associated with the conversation are cleared.

On top of that, user-provided feedback and corrections shape the system’s behavior allowing it to adapt to the user’s preferences. When there is a need for memoization, the system stores both the previous query (the *corrected*) and the current query (the *correctee*) in the user’s namespace, in the database.

TODO: parler que store parallèle ?? **TODO: parler quelque part d'async??** **TODO: schéma avec flux entiers, database, ...** **TODO: décire que local database redis...** **TODO: mettre au format UML??** **TODO: parler au dessus coûte cher de faire un appel LLM**

An assumption still lies in the nature of the field *location* as it is assumed to either be set or unset. A set location does not necessarily imply that it is a valid municipality, inscribed in the published Swiss official

⁷<https://openai.com/>

commune register⁸. A solution is proposed in the section Geocontext Retriever 4.2.1.3.

TODO: dire que utiliser un dictionnaire difficile car mises à jour fréquentes par exemple ici ou plus bas ??

Finally, the query is routed according to the Figure 2:

- If clarification is needed (either because the need for clarification is explicitly requested, or fields are missing), the request is sent to the clarify query agent (2).
- If the conversation type is a correction request, the query is sent to the geocontext retriever agent (4).
- If the intent is said to be actionable, the request is sent to both the geocontext retriever and the guidelines retriever agents (4,5).
- Otherwise, the query is sent to the geocontext retriever agent (4).

With the aim of the user's query now clearly defined, the next step is to address any ambiguities or missing information with the clarification agent.

4.2.1.2 – Clarify Query

Clarifying and resolving vagueness in the user's query is essential to better understand the fundamental intent and provide an aligned response.

With the output of the intent router agent properly defined, the two cases which lead to the need for clarification are either an explicit request for clarification due to ambiguity or missing information.

Those two cases are both handled at once as a language model is prompted with the user's query and missing fields to generate and stream a response inquiring for further information or clarification (Figure 2, transition 3). In the following turn, the newly provided information is merged with the previously deduced intent as designed and presented in the section Intent Router 4.2.1.1.

With no *structural* ambiguity left in the user's query, the intent router agent can now proceed to route the query to the **Geocontext Retriever** and **Guidelines Retriever** agents.

TODO: filer le prompt

4.2.1.3 – Geocontext Retriever

4.2.1.4 – Guidelines Retriever

Each layer is composed of modular components that interact through well-defined interfaces, ensuring flexibility and ease of maintenance. When the user prompts the system from the web interface, the query is routed to the AI agent in the backend. The AI agent will make use of two datasources: a local database (in the backend) and third-party APIs (in the external services layer).

Energy planning guidelines all come in a single Portable Document Format (PDF) and are available in french or german. They are broken down into multiple sources:

The primary document called *Vision 2060 et objectifs 2035* has been adopted in 2019 and sets intermediate targets for 2035 that take into account the natural resources of Valais/Wallis, current knowledge, as well as federal energy and climate policies with the ultimate goal of achieving a 100% renewable and indigenous energy supply. Moreover, the *Plan directeur 2019* adopted by the federal council on the 1st of May 2019, states the strategy for the canton's territorial development in the form of 49 information sheets, distributed across the five areas of activity: Agriculture, forest, landscape and nature; Tourism and leisure; Urbanization; Mobility and transport infrastructure; Supply and other infrastructure. Finally, the legal framework is defined by two key legislative documents. Notably, the *RS 705.1 - Loi sur les constructions (LC)* establishes the regulations for construction activities, while the *RS 730.1 - Loi sur l'énergie (LcEne)* defines the objectives and requirements for sustainable energy supply. TODO: citer autrement?

These documents are specifically designed and structured to convey information to the public. They are organized into sections, subsections or paragraphs which reference figures, tables, plots, past paragraphs

⁸<https://www.bfs.admin.ch/bfs/en/home/basics/swiss-official-commune-register.html>

and so on. Visual structure does not necessarily imply a logical flow of information or semantic structure. A document can look organized but still lack a proper machine readable structure. In practice, it is neither realistic nor scalable to expect a human to manually extract all the key information needed for energy planning from such complex documents. Therefore, it becomes essential to delegate this task to the computer, enabling automated extraction and processing of documents.

When data lack clear structure, it becomes difficult to extract information using algorithms or systematic procedures. However, advances in Multimodal Large Language Models (MLLMs) offer a solution as these models are designed to process and understand information presented in various modalities such as text, images, audio, and video. Paired with existing methods that are able to extract raw text from these documents, it has become easier to extract precise information from visually organized and heterogeneous documents by understanding not only the way information is displayed but also its underlying semantic meaning.

As such, a systematic approach is applied when extracting information from these documents:

- Raw text is extracted from the documents on a per-page basis.
- Each and every page is rendered into an image.
- Rendered pages and associated text are processed using MLLMs to deliver a translated interpretation and summary of the information contained within the page.

The extracted information is formatted in markdown, utilizing headings to structure the summary. It is then broken down into smaller chunks, each chunk being a “chapter” derived from the markdown content. Since only individual chunks are considered in subsequent steps, there is no need to perform an analysis across neighboring pages to ensure that the information is retrieved from its full context.

Finally, the extracted information is encoded into a vector representation -an embedding- and stored in the local database along with its associated chunks and metadata. An embedding is a mathematical representation of data in a high-dimensional vector space where semantically similar information are mapped to nearby points. This enables the system to embed text queries and efficiently retrieve semantically relevant information when compared with the stored embeddings.

TODO: cite plus tard que Retrieve multiple chunks recreate the context TODO: reformuler et enlever les formulations 'we' ? TODO: citer modèle utilisé MLLM et embeddings ? TODO: citer comment traiter données communales TODO: citer prompt TODO: inclure prompts dans bibliographie

4.2.1.5 – Strategy Planner

4.2.1.6 – Critic

4.2.2 – Web Interface

4.2.3 – Limitations

Chapter 5 – Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiat earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignorance rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatumque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus,

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Chapter 6 – Discussion

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Chapter 7 – Conclusion

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ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitatem aliqua naturam ipsam movet et cum iucunditate quadam percipitur sensibus, sed maximam voluptatem illam habemus, quae percipitur omni dolore careret, non modo non repugnantibus, verum etiam approbantibus nobis. Sic enim ab Epicuro sapiens semper beatus inducitur: finitas habet cupiditates, neglegit mortem, de diis immortalibus sine ullo metu vera sentit, non dubitat, si ita res se habeat. Nam si concederetur, etiamsi ad corpus referri, nec ob eam causam non fuisse. – Torquem detraxit hosti. – Et quidem se texit, ne interiret. – At magnum periculum adiit. – In oculis quidem exercitus. – Quid ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspicitur potest: Constituamus aliquem magnis, multis, perpetuis fruenter et animo et attento intuemur, tum fit ut aegritudo sequatur, si illa mala sint, laetitia, si bona. O praeclaram beate vivendi et apertam et simplicem et directam viam! Cum enim certe nihil homini possit melius esse quam Graecam. Quando enim nobis, vel dicam aut oratoribus bonis aut poetis, postea quidem quam fuit quem imitarentur, ullus orationis vel copiosae vel elegantis ornatus defuit? Ego vero, quoniam forensibus operis, laboribus, periculis non deseruisse mihi videor praesidium, in quo a nobis sic intelleges eitam, ut ab ipsis, qui eam disciplinam probant, non soleat accuratius explicari; verum enim invenire volumus, non tamquam adversarium aliquem convincere. Accurate autem quondam a L. Torquato, homine omni doctrina erudito, defensa est Epicuri sententia de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae praeterierunt, acri animo et corpore voluptatibus nullo dolore nec impediante nec inpendente, quem tandem hoc statu praestabiliorem aut magis expetendum possimus dicere? Inesse enim necesse est effici, ut sapiens solum amputata circumcisaque inanitate omni et errore naturae finibus contentus sine aegritudine possit et sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspicitur potest: Constituamus aliquem magnis, multis, perpetuis fruenter et animo et corpore voluptatibus nullo dolore nec impediante nec inpendente, quem tandem hoc statu praestabiliorem aut magis expetendum possimus dicere? Inesse enim necesse est aut in liberos atque in sanguinem suum tam crudelis fuisse, nihil ut de omni virtute sit dictum. Sed similia fere dici possunt. Ut enim virtutes, de quibus neque depravate iudicant neque corrupte, nonne ei maximam gratiam habere debemus, qui hac exaudita quasi voce naturae sic eam firme graviterque comprehenderit, ut omnes bene sanos ad iustitiam, aequitatem, fidem, neque homini infanti aut inpotenti.

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TODO: mettre une deuxième bib. technique uniquement!!

Appendix

Table of Acronyms

AI :	Artificial Intelligence
API :	Application Programming Interface
FSM :	Finite-State Machine
LLM :	Large Language Model
MLLM :	Multimodal Large Language Model
PDF :	Portable Document Format
RAG :	Retrieval-Augmented Generation
RLM :	Reasoning Language Model

List of Figures

Figure 1 Global system design	21
Figure 2 AI agent architecture	23